Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

How many ions are produced from $[Co(NH_3)_6]Cl_3$ in the solution?

(A) 3

(B) 4

(C) 5

(D) 6

Q.2

Which of the following statements is not correct?

(A) Primary valency of the central metal of a complex is always satisfied by anions

(B) Secondary valency of the central metal of a complex may be satisfied by either negative ions or neutral molecules

(C) Species attached by primary valencies in a complex compound can be precipitated out

(D) Species attached through secondary valencies in a complex are Lewis acids

Q.3

The complex $[Ni(NH_3)_6]^{2+}$ involves

(A) sp^3 hybridization

(B) dsp^3 hybridization

(C) d^2sp^3 hybridization

(D) $sp^3\ d^2$ hybridization

Q.4

The IUPAC name of $[Pt(NH_3)_4(NO_2)Cl]SO_4$ is

(A) tetramminechloronitroplatinum(II) sulphate

(B) tetramminechloronitroplatinum(IV) sulphate

(C) chlorotetrammine nitroplatinum(IV) sulphate

(D) chloronitrotetrammineplatinum(IV) sulphate

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Q.5

Which of the following complexes exhibits a magnetic moment of 1.73 bohr magneton?

(A) $[\mathrm{Cu}\,(\mathrm{NH_3})_4]^{2+}$

(B) $[\mathrm{NiCl_4}]^{2-}$

(C) $\mathrm{TiCl_4}$

(D) $[\mathrm{CoCl_6}]^{4-}$

Q.6

The compound $[\mathrm{Co}\,(\mathrm{NH_3})_2\ (\mathrm{en})\ \mathrm{Cl_2}]$ can form

(A) linkage isomers

(B) coordination isomers

(C) optical isomers

(D) linkage as well as optical isomers

Q.7

The complex $[\mathrm{Pt(py)}\,(\mathrm{NH_3})\,(\mathrm{NO_2})\,\mathrm{ClBrI}]$ has

(A) 10 geometrical isomers, each is optically active

(B) 10 geometrical isomers, five of them are optically active

(C) 15 geometrical isomers, each is optically active.

(D) 15 geometrical isomers, each is optically inactive.

Q.8

Vitamin $\mathrm{B_{12}}$ is a complex of

(A) cobalt(II) ion

(B) cobalt(III) ion

(C) chromium(II) ion

(D) chromium(III) ion

Q.9

Which of the following statements is not correct?

(A) Weak ligands like $\mathrm{F^-}\,\mathrm{Cl^-}$ form high-spin complexes

(B) Strong ligands like $\mathrm{CN^-}$ form low-spin complexes

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(C) $[FeF_6]^{3-}$ is a high-spin complex

(D) $Ni(CO)_4$ is a high-spin complex

Q.10

The IUPAC name of the complex having formula $[(CO)_3Fe(CO)_3Fe(CO)_3]$ is

(A) Monocarbonylferrate(0)

(B) Tricarbonyliron (0) – μ-tricarbonyliron(0)

(C) Tri- μ-carbonylbis- $\{$ tricarbonyliron $(0)\}$

(D) Hexacarbonyl- μ-tricarbonyliron(III)

Q.11

The IUPAC name for $[Pt(Br)(Cl)\,(NH_3)_3\,(NO_2)]\,Cl$ is

(A) Triamminechloridobromidonitroplatinum(IV) chloride

(B) Triamminebromidochloridonitroplatinum(IV) chloride

(C) Triamminnitrochlorobromoplatinum(IV) chloride

(D) Triamminechloronitrobromoplatinum(IV) chloride

Q.12

The number of donor sites in dimethyl glyoxime, glycinato, diethylene triamine and EDTA are respectively:

(A) $2, 2, 3$ and 4

(B) $2, 2, 3$ and 6

(C) $2, 2, 2$ and 6

(D) $2, 3, 3$ and 6

Q.13

$0.001\ mol$ of $Cr\,(NH_3)_5\,(NO_3)\,(SO_4)$ was passed through a cation exchanger and the acid coming out of it required $20\ mL$ of $0.1MNaOH$ for neutralisation. Hence, the complex is

(A) $[Cr\,(NH_3)_5\,SO_4]\,NO_3$

(B) $[Cr\,(NH_3)_5\,NO_3]\,SO_4$

(C) $[Cr\,(NH_3)_5]\,(SO_4)\,(NO_3)$

(D) None of these

Q.14

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EAN of $Na\left[PtCl_3\left(\eta^2-C_2H_4\right)\right]$ is

(A) 86

(B) 78

(C) 84

(D) 34

Q.15

The following complexes are given
(1) trans- $\left[Co\left(NH_3\right)_4 I_2\right]^{\oplus}$
(2) cis- $\left[Co\left(NH_3\right)_2(en)_2\right]^{3+}$
(3) trans- $\left[Co\left(NH_3\right)_2(en)_2\right]^{3+}$
(4) $\left[NiI_4\right]^{2-}$
(5) $\left[TiF_6\right]^{2-}$
(6) $\left[CoF_6\right]^{3-}$
Choose the correct code.

(A) 4,5 are coloured; 6 is colourless

(B) 2 is optically active; 1,3 are optically inactive

(C) 1, 2 are optically active; 3 optically inactive

(D) 4 is coloured; 5,6 are colourless

Q.16

In which of the following pairs both the complexes shoy optical isomerism?

(A) $cis-\left[Cr\left(C_2O_4\right)_2 Cl_2\right]^{3-}$, cis- $\left[Co\left(NH_3\right)_4 Cl_2\right]$

(B) $\left[Co(en)_3\right] Cl_3$, $cis-\left[Co(en)_2Cl_2\right] C]$

(C) $\left[Co\left(NO_3\right)_3\left(NH_3\right)_3\right], cis-\left[Pt(en)_2Cl_2\right]$

(D) $[PtCl(en)Cl]$, $\left[NiCl_2Br_2\right]^{2-}$

Q.17

Which of the following organometallic compound is a σ and π-bonded?

(A) $\left[Fe\left(\eta^5-C_5H_5\right)_2\right]$

(B) $K\left[PtCl_3\left(\eta^2-C_2H_4\right)\right]$

(C) $\left[Co(CO)_5NH_3\right]^{2+}$

(D) $\left[Fe\left(CH_3\right)_3\right]$

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Q.18

$[Cr(H_2O)_6]Cl_3$ (atomic number of $Cr = 24$) has a magnetic moment of $3.83 BM$. The correct distribution of $3d$-elections in the chromium present in the complex is

(A) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$

(B) $3d^1_{xy}, 3d^1_{yz}, 3d^1_{z^2}$

(C) $3d^1{}_{(x^2-y^2)}, 3d^1_{z^2}, 3d^1_{zx}$

(D) $3d^1_{xy}, 3d^1_{(x^2-y^2)}, 3d^1{}_{xz}$

Q.19

Which of the following statements is correct for the complex $Ca_2[Fe(CN)_5O_2]$ having $t^6_{2\,g}, e^0_g$ electronic configuration?

(A) d^2sp^3 hybridised and diamagnetic

(B) sp^3d^2 hybridised and paramagnetic

(C) sp^3d^2 hybridised and diamagnetic

(D) d^2sp^3 hybridised and paramagnetic

Q.20

Select the correct order of C-O bond order in mixed phosphine carbonyl complex:
I. $[Mo(Ph_3P)_3(CO)_3]$
II. $[Mo(Ph_2PCl)_3(CO)_3]$
III. $[Mo(PhPCl_2)_3(CO)_3]$

(A) $I > II > III$

(B) $I > II < III$

(C) $I = II > III$

(D) $I < II > III$

Q.21

The number of geometrical isomers of $[Cr(NH_3)(OH)_2Cl_3]^{2-}$ ion is

Q.22

The total number of possible isomers of $[CrCl(CN)(NH_3)$ en $]$ is

Q.23

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Coordination number of Cu^{2+} in $CuSO_4 \cdot 5H_2O$ is

Q.24

Give the number of ligands which are monodentate as well as ambidentate.

$\overset{\Theta}{C}N, C_2O_4^{2-}, S_2^{2-}, NO_2^{\Theta}, \overset{\Theta}{O}CN, SCN^{\Theta}$

Q.25

Predict the sum of unpaired electrons in a tetrahedral d^6 ion and in a square planar d^7 ion.

Q.26

How many maximum atom(s) is/are present in same plane of $Cr(CO)_6$?

Q.27

For the complex $[Cr\ (H_2O)_6]^{2+}$, calculate the total pairing energy in high spin and low spin state. Given the mean pairing energy $= 23500\ cm^{-1}$

Q.28

The sum of primary and secondary valencies of chromium in the complex $CrBr_3 \cdot 6NH_3$ is:

Q.29

What are the sum of the values of m and n in the anionic species $V(CO)_m]^{n-}$, if it is following Sidwick EAN rule and having octahedral shape?

Q.30

Consider the following carbonyl complex compounds
(I) $H_xCr(CO)_5$
(II) $CO(CO)_y$
(III) $Mo(CO)_z$
then calculate the value of $x + y + z$ is:

Answers & Solutions

Q.1 Answer:

4

Solution:

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4 ions are produced from $[Co(NH_3)_6]Cl_3$ in solution. Out of which first ion is $[Co(NH_3)_6]^{3+}$ and second ion is $3Cl^-$.

Q.2 Answer:

Species attached through secondary valencies in a complex are Lewis acids

Solution:

Species attached through secondary valencies in a complex are Lewis bases.

Q.3 Answer:

sp^3d^2 hybridization

Solution:

$Ni^{2+}(3d^8)$ ion in the given complex involves sp^3d^2 hybridization

Q.4 Answer:

tetramminechloronitroplatinum(IV) sulphate

Solution:

Given the compound, $[Pt(NH_3)_4(NO_2)Cl]SO_4$.
Here, we know SO_4 has -2 charge.
So, the overall charge on $[Pt(NH_3)_4(NO_2)Cl]$ is $+2$.Cl and NO_2 has -1 valency each.
Therefore, charge on Pt has to be $+4$ to balance the overall charge.
Therefore, the IUPAC name of $[Pt(NH_3)_4(NO_2)Cl]SO_4$ is tetramminechloronitroplatinum (IV) sulphate.

Q.5 Answer:

$[Cu(NH_3)_4]^{2+}$

Solution:

1.73 Bohr magneton implies one unpaired electron according to the expression $\sqrt{n(n+2)}\mu_B$.
The species Cu, Ni, Ti and Co in the complexes have $(3d)^9, (3d)^8, (3d)^0$ and $(3d)^7$ electronic configurations, respectively. This involve sp^2d, dsp^2, sp^3 and d^2sp^3 hybridization, respectively.
Thus, the complex $[Cu(NH_3)_4]^{2+}$ has one unpaired electron.

Q.6 Answer:

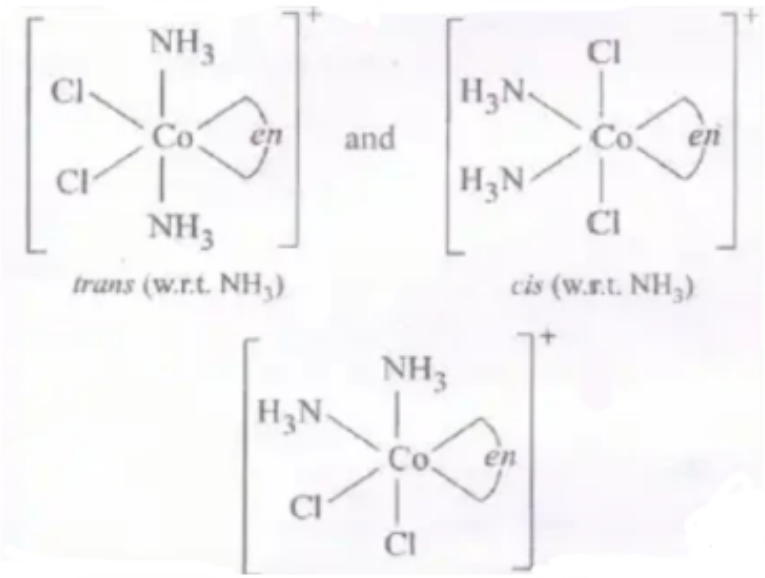
optical isomers

Solution:

The compound $[CoCl_2(NH_3)_2(en)]$ has two geometrical isomers. One is the cis-isomer (with respect to NH_3) and the other is the trans- isomer (with respect to NH_3).
The cis-isomer has two optically active forms, d-cis isomer and l-cis isomer. So, the compound

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$[CoCl_2(NH_3)_2$ **(en)] can form geometrical isomers as well as optical isomers.**



Q.7 Answer:

15 geometrical isomers, each is optically active.

Solution:

The complex [Mabcdef] has 15 geometrical isomers and all of them are optically active.

Q.8 Answer:

cobalt(III) ion

Solution:

Vitamin B_{12} is a complex of cobalt. It is cyanocobalamin $(C_{63}H_{88}CoN_{14}O_{14}P)$. Its main sources are meat, fish, egg, milk etc. Its deficincy is responsible for pernicious anaemia.

Q.9 Answer:

$Ni(CO)_4$ is a high-spin complex

Solution:

F^- and Cl^- are weak ligands. These have small tendency of pairing metal electrons. Thus, these form high-spin complexes. The spacing of splitted orbitals of metal ions is small.
CN^- is a strong ligand. It has high tendency of pairing metal electrons. Thus, it forms low-spin complexes. CO is a strong ligand and thus it forms low-spin complexes.

Q.10 Answer:

Tri- μ-carbonylbis- $\{$ tricarbonyliron $(0)\}$

Solution:

The bridging ligands which link the two metal atoms together are indicated by the prefix $-\mu-$. If there are two or more bridging groups of the same kind, this is indicated by $di-\ \mu-$, $tri-\ \mu-$ etc. As carbonyl is neutral ligand oxidation number of metal (Fe) will be zero.

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Q.11 Answer:

Triamminebromidochloridonitroplatinum(IV) chloride

Solution:

We need to follow the alphabetical order while naming the compound. The oxidation number of metal is $+4$.

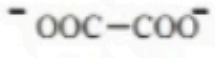
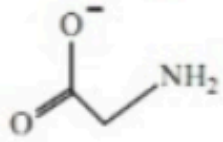
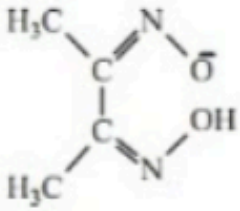
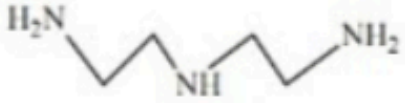
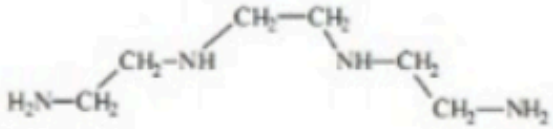
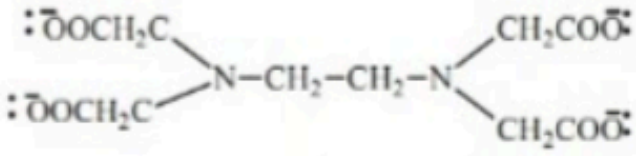
Triamminebromidochloridonitroplatinum(IV) chloride

Q.12 Answer:

$2, 2, 3$ and 6

Solution:

2,2,3 and 6

Oxalate	Ox(Bidentate)	
Glycinato	gly(Bidentate)	
Dimethylglyoximate	dmg	
Diethylenetriamine	dien	
Triethylenetetramine	tren	
Ethylendiamine tetraacetate	E.D.T.A.	 (Hexadentate)

Q.13 Answer:

$[\mathrm{Cr}\,(\mathrm{NH_3})_5\,\mathrm{NO_3}]\,\mathrm{SO_4}$

Solution:

The complex is $[\mathbf{CoNO_3\,(NH_3)_5}]\,\mathbf{SO_4}$

$\frac{0.020\ \mathrm{L}\times 0.1\ \mathrm{mol/L NaOH}\times 1\ \mathrm{mol\ acid}}{1\ \mathrm{mol NaOH}} =$

0.0020 **mol acid**

$\frac{0.0020\ \mathrm{mol}}{10^{-3}\ \mathrm{mol}} = 2$

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As 2 moles of acid in the exchanger replaced each 1 mole of complex cations, so the complex must have an oxidation state of $+2$, so the complex will be $\left[\mathrm{CoNO_3\left(NH_3\right)_5}\right] \mathrm{SO_4}$.

Q.14 Answer:

84

Solution:

$\mathrm{K}\left[\mathrm{PtCl_3}\left(\eta^2-\mathrm{C_2'H_4}\right)\right]$
$+4$
$\mathrm{EAN}=74+3^*2+2*2=84$

Q.15 Answer:

2 is optically active; 1,3 are optically inactive

Solution:

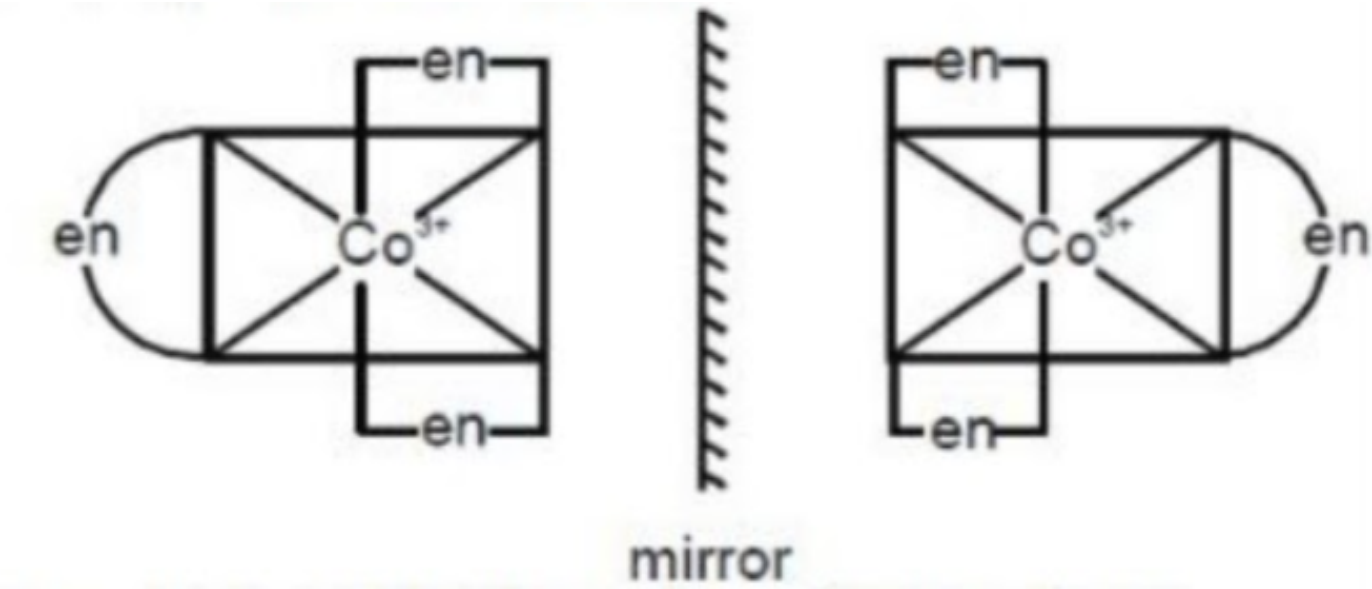
2 is optically active; 1,3 are optically inactive

Q.16 Answer:

$[\mathrm{Co(en)_3}]\ \mathrm{Cl_3}, \text{cis}-[\mathrm{Co(en)_2Cl_2}]\ \mathrm{C}]$

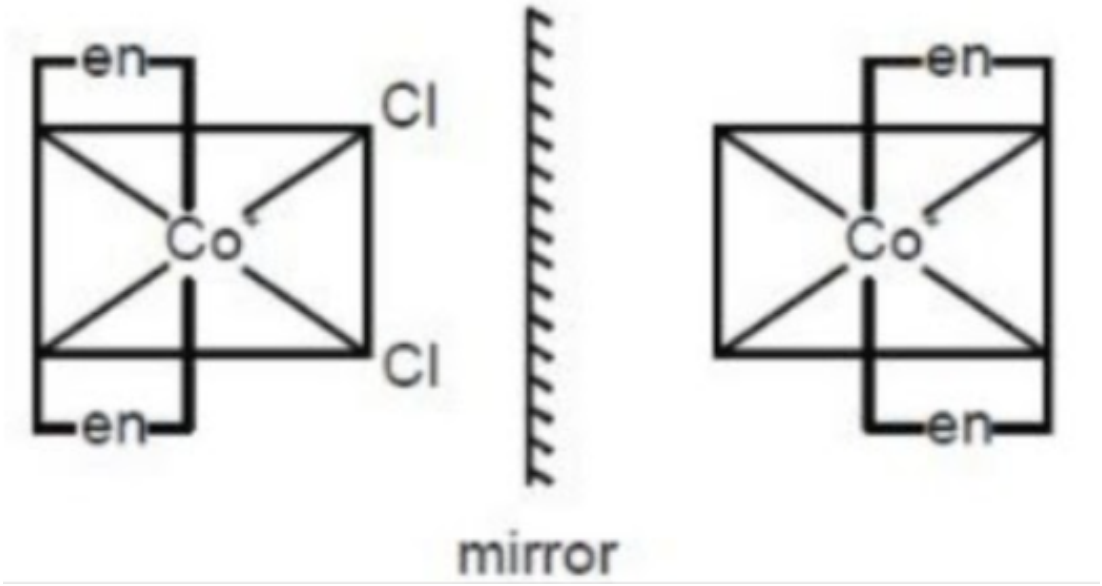
Solution:

$[\mathrm{Co(en)_3Cl_3}]$ **i.e.,** $[\mathrm{Co(en)_3}]^{3+}$



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cis $-[\mathrm{Co(en)_2Cl_2}]\,\mathrm{Cl}$ **i.e. cis** $-[\mathrm{Co(en)_2Cl_2}]^{2+}$



Q.17 Answer:

$[\mathrm{Co(CO)_5NH_3}]^{2+}$

Solution:

$[\mathrm{Co(CO)_5NH_3}]^{+2}$ **is** σ **and** π **bonded. In this organometallic compound,bond between cobalt and CO ligands have both** σ **and** π **character.** σ**-bond formation between metal and carbon of CO takes place by overlap of filled bonding** $\pi_{2\mathrm{p}}$ **of CO with an empty metal** d **- orbital. The 2nd overlap takes place between metal d- orbital with an empty antibonding pi orbital of the** CO **resulting in additional** $\pi-$ **bond between the metal and same CO molecule.**

Q.18 Answer:

$3d^1_{xy}, 3d^1_{yz}, 3d^1_{zx}$

Solution:

$[\mathrm{Cr}\,(\mathrm{H_2O})_6]\,\mathrm{Cl_3}$ **(at, no. of** $\mathrm{Cr} = 24$ **) has a magnetic moment of** 3.83 **B.M. The correct distribution of 3 d electrons of chromium in the complex is** $3\ \mathrm{d}^1_{\mathrm{xy}}, 3\ \mathrm{d}^1_{\mathrm{yzz}}, 3\ \mathrm{d}^1_{\mathrm{xz}}$.

The magnetic moment of 3.83 **B.M corresponds to 3 unpaired electrons.**

$$3.83 = \sqrt{\mathrm{n}(\mathrm{n}+2)}$$
$$\mathrm{n} = 3$$

All these electrons are present in $\mathrm{t}_{2\,\mathrm{g}}$ **orbitals. These correspond to** $3\ \mathrm{d}^1_{\mathrm{xy}}, 3\ \mathrm{d}^1_{\mathrm{yz}}, 3\ \mathrm{d}^1_{\mathrm{xz}}$.

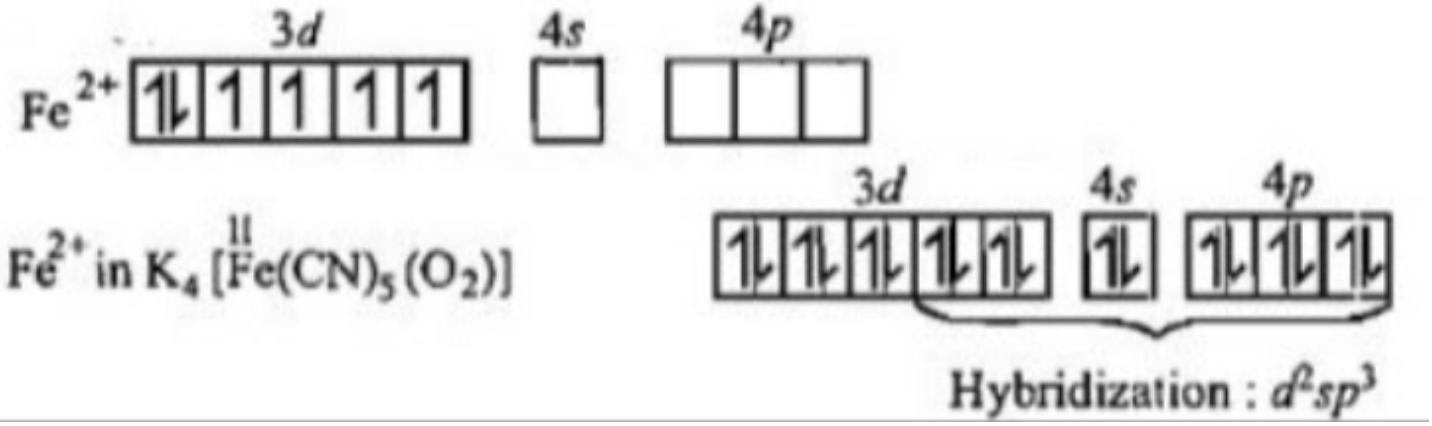
Q.19 Answer:

d^2sp^3 hybridised and diamagnetic

Solution:

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Complex is dimagnetic due to presence of paired electron at O_2^- - i.e., superoxide acting as ligand.



Q.20 Answer:

$I > II < III$

Solution:

As phenyl group increases, resonance increases and bond length increases so order is

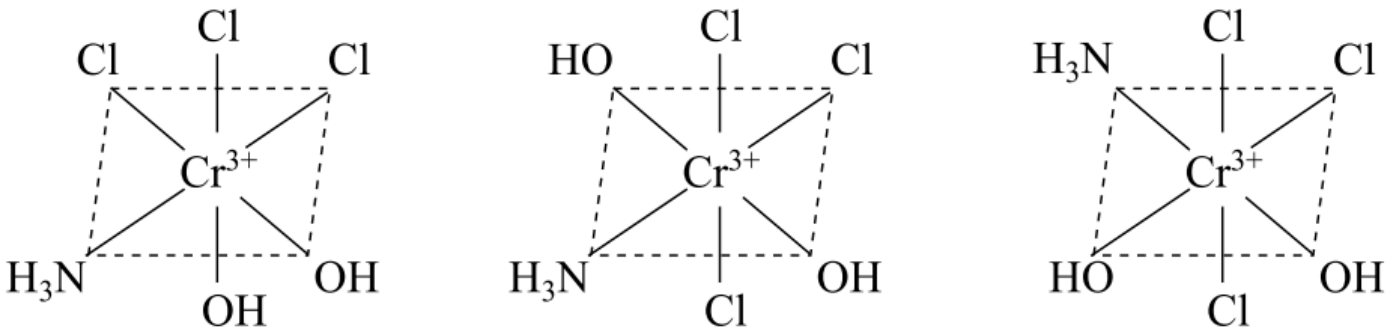
$[(Ph_3P)_3\,Mo(CO)_3] > [(Ph_2PCl)_3\,Mo(CO)_3] < [(PhPCl_2)_3\,Mo(CO)_3]$

Q.21 Answer:

—

Solution:

There are three geometrical isomers:



cis Cl—Cl
cis OH—OH

trans Cl—Cl
trans OH—OH

trans Cl—Cl
cis OH—OH

Q.22 Answer:

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Solution:

The complexes of coordination chemishy, mostly have numbers of isomers. The isomers can be geometrical or stereoisomer.

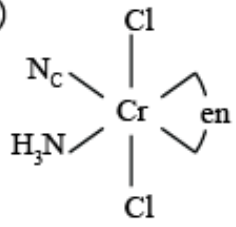
Given: $[CrCl(CN)\,(NH_3)$ **en** $]$

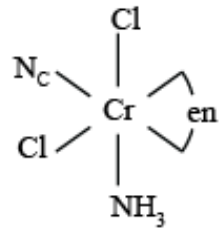
denticity of en iS 2, thus coordination. no of Cr **here is 6 and the oxidation state can be calculated** $as \rightarrow$

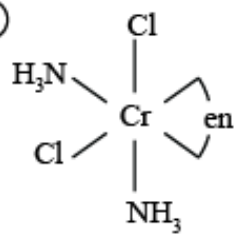
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$\Rightarrow x + (-2) + (-1) + 0 + 0 = 0$

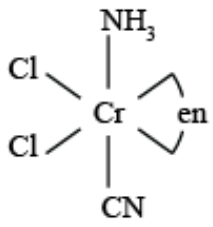
$\Rightarrow x = +3$

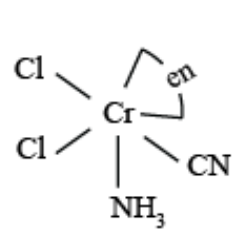
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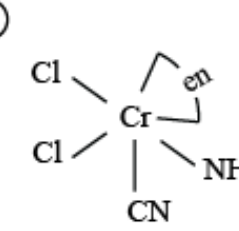
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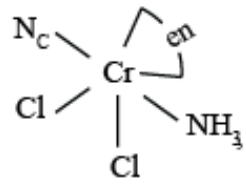
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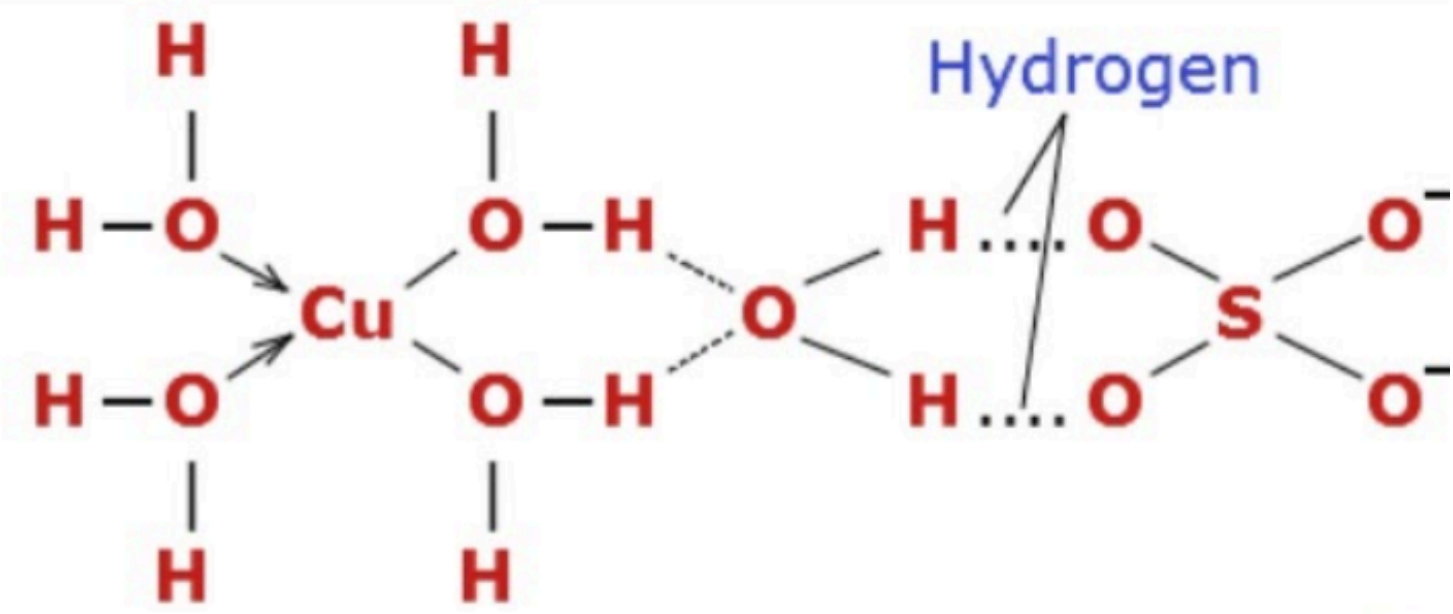
$\therefore$ **Possible isomer** $= 7$

Q.23 Answer:

—

Solution:

Coordination number of Cu^{2+} **is 4**



Q.24 Answer:

—

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Solution:

Ambidentate ligands have 2 possible donor site

$\rightarrow$ They can coordinate through single atom at a time

Total $=4$ which are

$\ominus \quad \ominus \quad \ominus \quad \ominus$
$CN, NO_2, OCN, \text{and } SCN$

Q.25 Answer:

—

Solution:

The number of unpaired electrons in a tetrahedral d^6 ion is 4

The number of unpaired electrons in square planar d^7 ion is 1 .

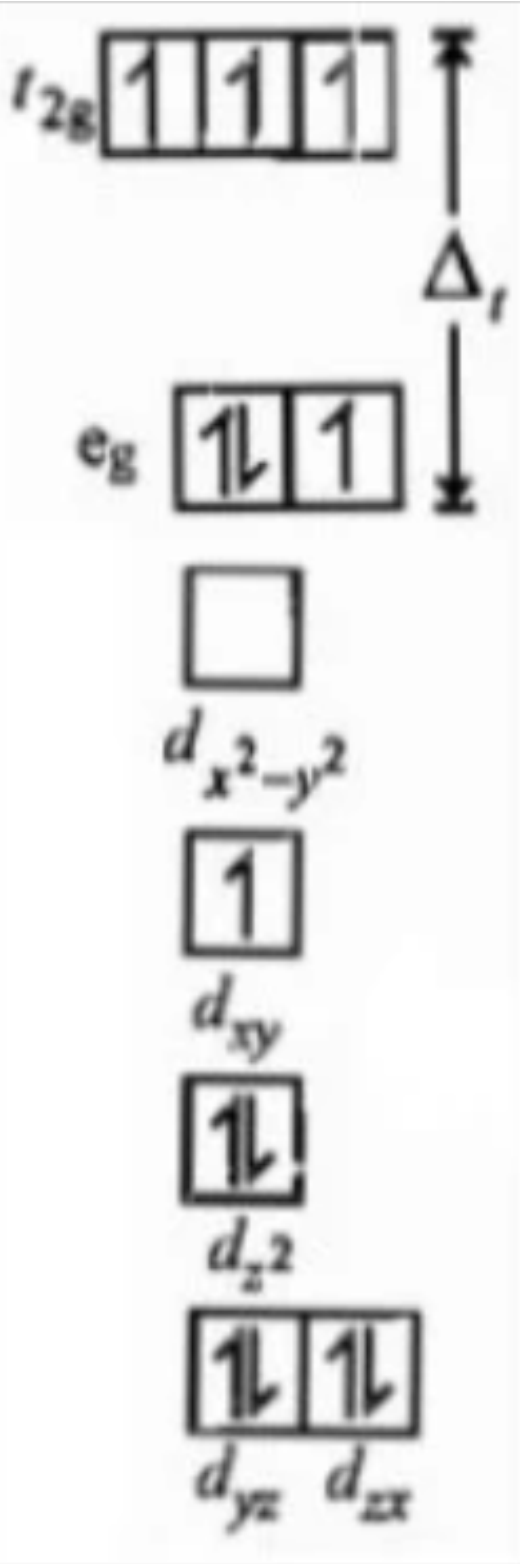
Electronic distribution in a tetrahedral d^6 ion is $\mathrm{e}_{\mathrm{g}}^3 \mathrm{t}_{2\,\mathrm{g}}^3$

Number of unpaired electrons is 4

Electronic distribution in square planar d^7 ion $d_{yz}^2 d_{zx}^2$ $\mathrm{d}_{z^2}^2$ $\mathrm{d}_{xy}^1 d_{x^2-y^2}^0$

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Number of unpaired electrons is 1

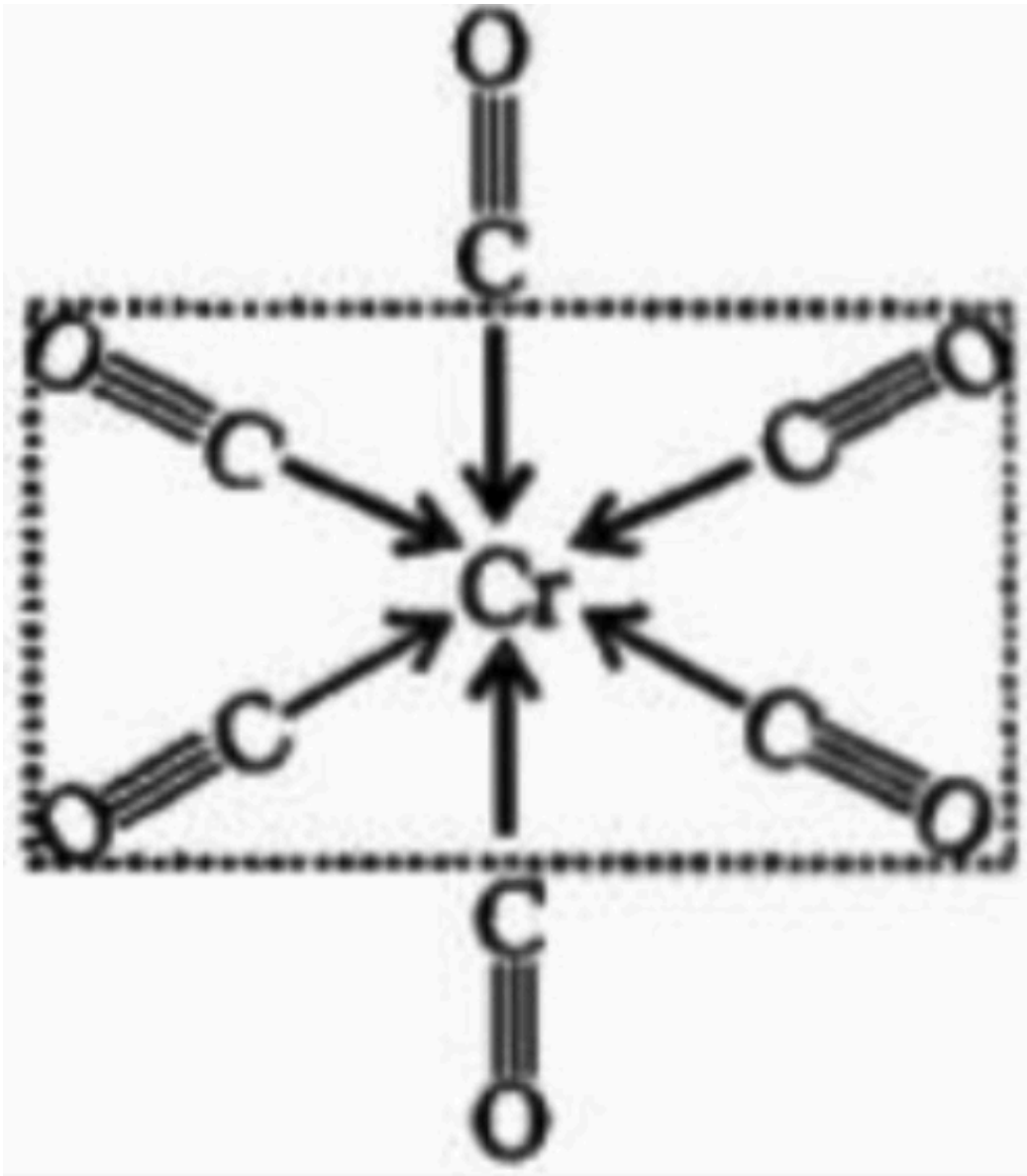


Q.26 Answer:

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Solution:

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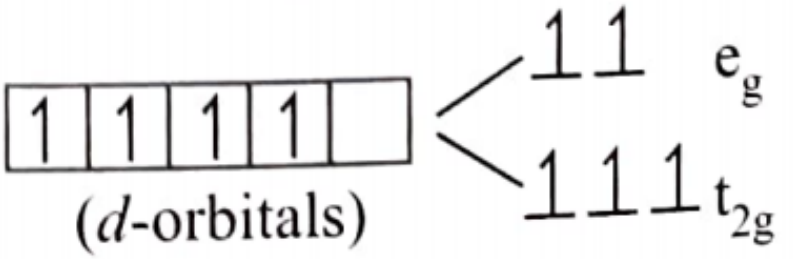


Q.27 Answer:

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Solution:

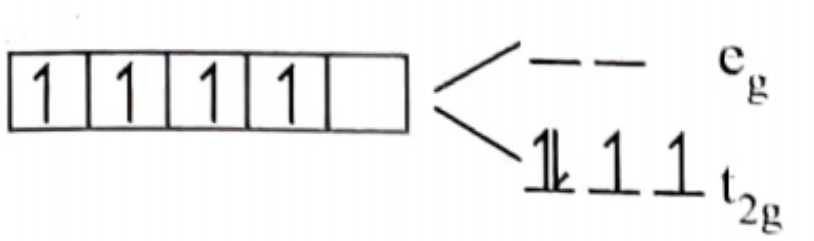
In complex $[\mathrm{Cr}\left(\mathrm{H_2O}\right)_6]^{2+}$, oxidation state of Cr is $+2$, i.e. Cr^{2+} is present which has 3 d^4 configuration. The configuration of d^4 ion in high spin state is ${t_{2g}}^3 e_g'$, since no electron is paired, total pairing energy of $[\mathrm{Cr}\left(\mathrm{H_2O}\right)_6]^2$ in high spin state is $0 \times P = 0$.



The configuration of d^4 ion in low spin state is $\mathrm{t}_{2\,\mathrm{g}}^4 \mathrm{e}_\mathrm{g}^0$, since pairing of an electron takes place, total pairing energy of $[\mathrm{Cr}\left(\mathrm{H_2O}\right)_6]^{2+}$ in low spin state is $1 \times P = 1 \times 23500\ \mathrm{cm}^{-1} =$

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$23500\ cm^{-1}$.



Q.28 Answer:

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Solution:

Primary valency = oxidation state secondary valency = Number of molecule co-ordinate to central atom

In the molecule $CrCl_3 \cdot 6NH_3$

Cr is $+3$ **state . Hence Primary valency** $= 3$

$six(6)NH_3$ **is co-ordinate with central atom.**

Hence, secondary valency $= 6$

Sum $= 3 + 6 = 9$

Hence, correct answer is 9 .

Q.29 Answer:

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Solution:

Octahedral shape corresponds to $6CO$ **ligands.**

Hence, $m = 6$

EAN $= 23 + n + 2 \times 6 = 23 + n + 12 = 35 + n = 36.$

Hence, $n = 1.$

Thus, the complex is $[V(CO)_6]^-$.

Q.30 Answer:

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Solution:

The carbonyls in general obey the effective atomic number rule and their effective atomic number = atomic number of next noble gas

(i) $Mo(CO)_x \Rightarrow EAN = 54'$

$42 + 2x = 54$

$x = 6$

(ii) $H_yCr(CO)_5 \Rightarrow EAN = 36'$

$24 + (2 \times 5) + y = 36$

$y = 2$

(iii) $Co_2(CO)_z \Rightarrow EAN = 36'$

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$27 + 1(\text{ from other Co-atom }) + 2z = 36$

$z = 4$

then,

$|x + y - z| = 6 + 2 - 4 = 4$

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